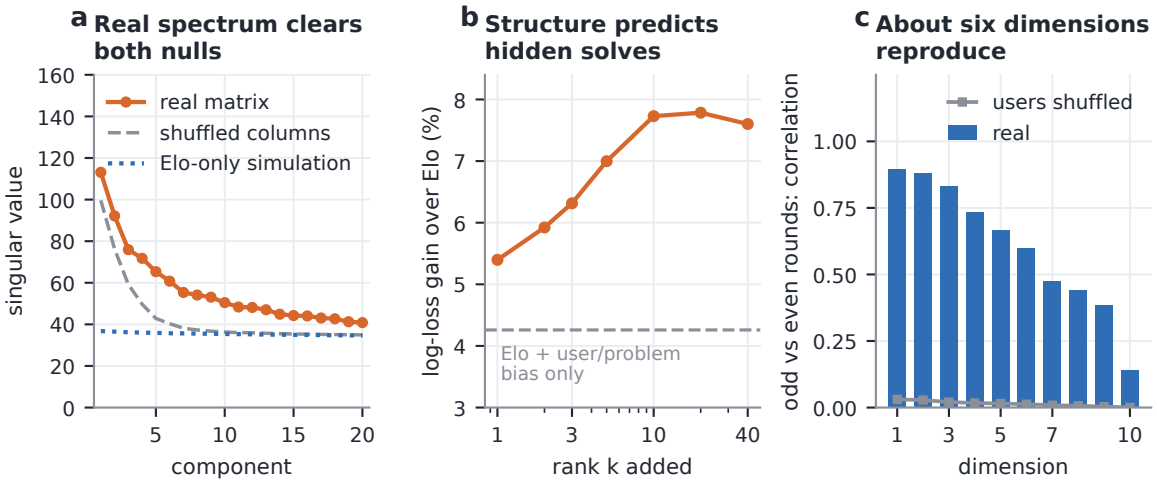
a
Real spectrum clears both nulls
real matrix
shuffled columns
Elo-only simulation
singular value
component
b
Structure predicts hidden solves
log-loss gain over Elo (%)
Elo + user/problem bias only
rank k added
c
About six dimensions reproduce
users shuffled
real
odd vs even rounds: correlation
dimension